[redacted] Solar Power Forecasting

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1. DATA & PREPROCESSING

1.1. Selection of physics-motivated features

The most significant input across the models outlined in this report are lagged power observations: naturally, recent power data is a strong indicator of future power generation. On short horizons this exploits temporal continuity directly, while on longer horizons it provides contextualisation for meteorological features to act upon.

The meteorological dataset included minute-resolution measurements of irradiance, rainfall, relative humidity, sea level pressure, temperature, visibility, and wind speed. Of these, only irradiance, relative humidity, and temperature were chosen on physical grounds. Irradiance is the primary driver of PV generation, temperature governs panel efficiency through material physics, and relative humidity can reduce panel efficiency through surface condensation whilst providing a temporally leading indicator of incoming cloud cover and irradiance reduction. In contrast, the remaining metrics were excluded as their physical effects on PV generation are either indirect or second-order, offering no significant predictive advantage over the selected features.

Lastly, solar position was introduced as a feature using the pvlib library, with solar zenith and azimuth used to inform the models of diurnal cycles as well as seasonality. These angles were cyclically encoded to preserve the continuity of angles: without this, a model treating angles as scalar values would consider 359° and 0° as maximally distant.

For the multi-station model, station-wise static features were extracted from the metadata file to enable generalisation. Station altitude, capacity, tilt, and azimuth were kept due to their physical significance: station altitude affects air mass and irradiance attenuation; capacity determines the relative magnitude of station output; and tilt and azimuth govern the angle of incidence of incoming sunlight. Panel azimuth was also cyclically encoded, whereas tilt required no such treatment as its range of $[0^o, 90^o]$ was inherently non-periodic.

1.2. Weather/PV Dataset Merging, NaN Handling

Station PV data was recorded at either hourly or quarter-hourly resolution. Therefore, meteorological and PV datasets were resampled to hourly intervals so that model pipelines were station-adaptable by ensuring a consistent resolution across all stations. This resampling used mean aggregation, as it preserves the physical interpretation of power and irradiance as time-averaged quantities. The datasets were then merged on their shared time column, such that each row contained a station's PV output at a timestamp, accompanied by the weather features. Any rows containing a NaN were dropped at this point.

1.3. Set Splitting and Normalisation

The composite NaN-dropped dataframe was then split into a $75\% - 15\% - 15\%$ training-validation-test split. This was performed chronologically in order to prevent data leakage — otherwise future observations would appear in the training set, artificially inflating performance metrics. Feature normalisation was fitted exclusively on the training partition, and applied to validation and test sets. This prevented statistical information from out-of-sample data influencing training.

For single-station models, MinMax scaling was applied to all features, mapping values onto the interval $[0, 1]$. PyTorch's LSTM implementation uses sigmoid and tanh activations internally within its gating mechanism, so keeping inputs within this range ensures these functions operate in their sensitive regions (avoiding saturation and impeded gradient flow).

For multi-station models, power was instead normalised by station capacity whilst meteorological and static features retained MinMax scaling. This effectively mapped power to the same $[0, 1]$ interval whilst resolving station size differences. This capacity factor represents fractional utilisation, enabling the model to learn a single unified output scale across all 60 stations.

1.4. Sliding Window Sequence Construction

This project used LSTMs to model power generation, as explained in §2.1, and thus required sequential inputs. These were constructed as sliding windows with an adjustable lookback length and prediction horizon, allowing multi-step prediction. The primary results of this report used a lookback of 48 hours and horizons of 24 hours for single-station LSTMs, whilst the generalised LSTM used a 24h lookback to predict the next hour.

Windows were constructed to avoid temporal gaps: a sequence was only created if all required timesteps were contiguous, meaning longer lookback and horizon combinations yielded fewer sequences. A stride of 1 was used in order to maximise the number of available sequences. Sliding windows were constructed independently within each train/-val/test partition to prevent data leakage across splits.

2. METHODS

2.1. Baseline Single-Station LSTM

PV power generation is an inherently non-linear sequential process with dynamics operating across multiple timescales, such as hourly cloud transients to diurnal and seasonal cycles. This motivates a recurrent architecture capable of capturing and retaining long-range dependencies whilst staying responsive to short-term variations. LSTMs are exactly designed for this: their gating mechanism allows selective retention of information over arbitrarily long sequences, whereas vanilla RNNs fail due to vanishing gradients. LSTM was preferred over the GRU alternative as its explicit separation of cell and hidden states allows long-horizon context and short-term variability to be modelled independently — a property aligned to the multi-timescale dynamics of PV generation.

Model hyperparameters were chosen through manual tuning based on validation loss: hidden size and number of layers were increased until no further improvement was observed, whilst dropout was increased in response to observed training-validation loss divergence. The baseline single-station LSTM was designed as a 2-layer stacked LSTM with hidden size of 64, where stacking enables the second layer to learn higher-order abstractions over the first. The final state was passed through a two-layer fully-connected (FC) head with an intermediate ReLU activation, providing additional nonlinear mapping capacity to project the LSTM representation into a multi-step output. This avoids error accumulation present in autoregressive models.

As for regularisation, dropout was set at 0.2 between LSTM layers and 0.1 inside the FC head: it prevents co-adaptation of hidden units by randomly zeroing activations during training, forcing each neuron to serve a distinct purpose. The higher rate was used to moderate the greater overfitting risk in the deeper, higher-parameter recurrent stack, whilst the lighter rate was used in the FC head to preserve more of the sequence representation built by the LSTM before the final output projection.

Training used MSE loss, as it penalises large errors quadratically. In an applied scenario such as grid management and energy forecasting, large prediction errors are disproportionately costly compared to small ones. The choice of MSE loss over MAE was made to reflect that. Furthermore, the earlier MinMax scaling of all features complements this approach: all features contribute on a consistent scale to the loss, preventing high-magnitude features from dominating. The Adam optimiser was selected for its per-parameter adaptive learning rates, which naturally accommodate the mixed gradient magnitudes across LSTM gates that would challenge a fixed learning rate optimiser such as SGD. Learning rate scheduling was applied through PyTorch's ReduceLROnPlateau, allowing for aggressive early learning and finer convergence at later stages without manual intervention. This halved the learning rate if validation loss didn't improve in 3 epochs, with an initial rate of 0.001 and a floor of 1.0×10^{-4} to prevent excessive reduction. To avoid overfitting, early stopping was used, with training ending if validation loss did not improve in 15 epochs. The model weights that achieved the lowest validation loss were saved and restored at the end of training.

2.2. Single-Station GA-VMD-LSTM Ensemble

Whilst LSTMs can capture temporal dependencies, PV power contains many layers of overlapping periodic components — the diurnal cycle, seasonality, and weather transients — that the LSTM must disentangle. Variational Mode Decomposition (VMD) offers a solution to this: it decomposes a signal into a set of K band-limited intrinsic mode functions (IMFs), each centred on a distinct frequency. VMD was preferred over Empirical Mode Decomposition (EMD) as its variational formulation explicitly constrains each IMF to a distinct frequency band, preventing the mode mixing that EMD's recursive sifting produces. Decomposing training, validation, and test partitions separately — in order to prevent data leakage — and fitting LSTMs on each of the IMFs has previously yielded lower forecasting error than standalone LSTM baselines on PV power data [1].

VMD hyperparameters, K and α (bandwidth penalty factor) — were optimised using a genetic algorithm (GA) with an envelope entropy as the fitness function, as used in [2]. GA was chosen for its natural handling of the mixed discrete-continuous search space in optimising (K, α). This was only performed on the training partition to avoid data leakage. For each candidate (K, α) pairing, the power signal was decomposed and the mean envelope entropy across IMFs computed — quantifying the compactness of each mode's energy distribution via its Hilbert envelope. Lower entropy reflects cleaner, better-separated modes, making it a natural unsupervised criterion for decomposition quality. The GA used a population of 30 candidates evolved over up to 50 generations with tournament selection, arithmetic crossover for α and integer crossover for K, Gaussian mutation, and elitism of 2. The process terminated early if no improvement was observed over 10 consecutive generations. The search was performed over the discrete interval $K \in [2, 8]$ and the continuous interval $\alpha \in [100, 3000]$. The lower bound of $\alpha = 100.0$ was set to prevent near-zero bandwidth penalties, which would relax the frequency constraints on each IMF to the point where modes absorb noise rather than capturing distinct periodic components — undermining the physical motivation for decomposition. For station SQ19 — on which the main single-station analysis is performed — the GA converged on a pairing of $(7, 100.0)$.

The IMFs found by the VMD using these hyperparameters reconstructed the original power input with an $R^2 = 0.998$. Since this does not account for the full signal, the $K + 1$ LSTMs were trained on these IMFs and a residual term. Training was performed with the same LSTM hyperparameters as with the baseline model, except for the hidden size and the number of layers used. Hand-tuning of these found the best performance when allocating 2 layers and a hidden size of 64 to the 4 lowest-frequency IMFs, and 1 layer and a hidden size of 32 to the higher frequency modes. This mirrors the information content of each IMF: low-frequency signals

such as seasonality, diurnal cycles, and gradual weather patterns, require the most model capacity to learn whilst higher-frequency IMFs capture rapid fluctuations. The second-lowest frequency ($K = 2$) IMF was granted a larger LSTM hidden size of 96 as it encoded the primary diurnal periodicity influenced by weather variability: the most complex forecasting target in this decomposition. The final PV generation forecast was obtained by summing the individual $K + 1$ IMF LSTM predictions.

2.3. Multi-Station Generalised LSTM

Lastly, the baseline single-station LSTM architecture was extended to accept and generalise over all stations with sufficient data by considering their metadata. Rather than concatenating static features every timestep, these features are encoded into an embedding via a small linear network that flows into a tanh activation function, bounding the embedding to $[-1, 1]$ and keeping it compatible with the LSTM's inner gate activations. This is then projected to initialise both the h_0 and c_0 states for all LSTM layers. This seeds the LSTM's memory with station identity before time-series data is provided. Therefore, the model begins each sequence pre-conditioned to the station's physical properties, and avoids conflating time-variant and time-invariant information.

The AdamW optimiser was chosen in place of Adam as its decoupled weight decay applies consistent regularisation across all parameters regardless of gradient magnitude. In contrast, Adam's adaptive scaling reduces regularisation on frequently-updated weights, which can be problematic in a larger, more complex multi-station setting. The learning rate floor was lowered to 1×10^{-5}, and scheduling patience was increased from 3 to 5 epochs. Dropout between LSTM layers was raised to 0.4 so as to further prevent overfitting, which is more likely with higher complexity datasets. Early stopping patience was also increased from 15 to 20 epochs, and remaining architecture — hidden size of 64 and 2 layers — was retained from the baseline model.

3. RESULTS

3.1. Single-Station Models

Table I displays RMSE, MAE, and R^2 metrics from the full 24-hour horizon predictions made by both the baseline LSTM and the GA-VMD-LSTM Ensemble. GA-VMD-LSTM far outperforms LSTM, with 53% aggregate RMSE reduction ($2383 \rightarrow 1121$), 51% MAE reduction ($1258 \rightarrow 618$), and an R^2 improvement from 0.808 to 0.957. The unexplained variance fraction ($1 - R^2$) was reduced from 19.2% to 4.3% — a 78% reduction in unexplained variance.

Figure 1 plots the time-evolution of R^2 across the prediction horizon. Whilst the LSTM's performance sharply drops from short horizons onwards, GA-VMD-LSTM maintains $R^2 > 0.98$ through t+8. By this point, LSTM performance already degraded to 0.79. After t+10, LSTM plateaus

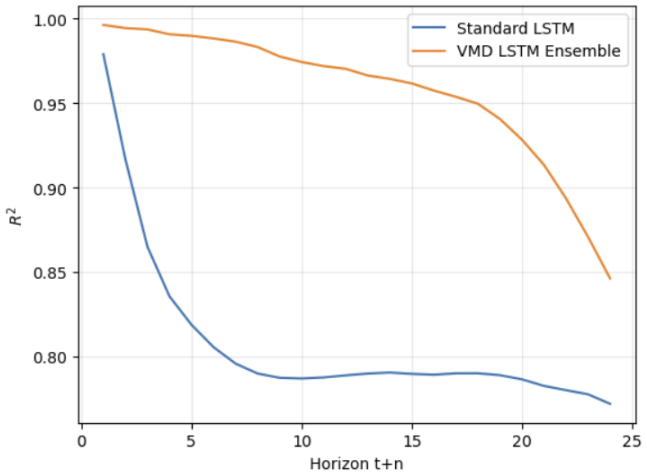


FIG. 1: Line plot of R^2 between model predictions and actual test set values across the full horizon for both the baseline LSTM and the GA-VMD-LSTM Ensemble. Modelling is of station SQ19.

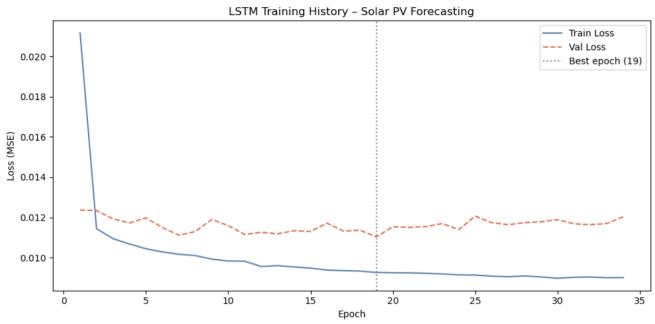


FIG. 2: Training and validation MSE loss curves per epoch for baseline LSTM. Vertical line at epoch 19 shows the best model checkpoint. Modelling is of station SQ19.

in the sub-0.8 range, suggesting the model loses temporal signals and converges to a safe mean prediction. In contrast, GA-VMD-LSTM degrades continuously and smoothly throughout the timeframe, suggesting healthy learning of physics. As the model looks further into the future, meteorological unpredictability dominates its error.

This is further supported by a comparison of the loss curves between models. Figure 2 shows the baseline LSTM's loss plateauing early whilst training loss continued to decrease — a signature of overfitting. In contrast, Figure 3 (and Appendix §7.1 for remaining IMFs) shows each IMF sub-LSTM achieving converging train and validation losses. This demonstrates that VMD successfully produces learnable sub-problems for each LSTM.

Individual IMF performance in Table II reveals that low-frequency components — IMFs 1-3, with $R^2 > 0.96$ — are highly predictable. The high-frequency IMFs yield substantially lower R^2, reflecting the random nature of short-term weather transients. The ensemble error is lower than what would be expected from the individual IMFs — demonstrating that their error is the result of unpredictable, phase-incoherent

	RMSE		MAE		R^2	
Horizon	LSTM	GA-VMD-LSTM	LSTM	GA-VMD-LSTM	LSTM	GA-VMD-LSTM
t+1	788.4955	333.3	481.4887	245.2	0.9789	0.9962
t+2	1565.9688	408.2	807.2961	291.4	0.9167	0.9943
t+3	1994.9806	435.2	1037.9517	311.2	0.8650	0.9936
t+4	2204.8330	525.3	1160.4579	365.8	0.8357	0.9907
t+5	2319.5800	551.2	1239.6127	382.5	0.8188	0.9898
t+6	2407.2559	592.6	1288.6237	417.0	0.8056	0.9882
t+7	2470.7262	639.4	1331.2260	441.8	0.7958	0.9863
t+8	2507.8725	710.2	1372.6642	473.5	0.7900	0.9832
t+9	2522.4843	819.8	1395.7534	531.8	0.7874	0.9775
t+10	2524.2460	877.2	1387.7169	566.9	0.7870	0.9743
t+11	2520.2588	917.5	1384.7684	582.9	0.7877	0.9719
t+12	2513.1813	943.2	1380.1271	595.3	0.7889	0.9703
t+13	2506.7960	1004.1	1351.9491	617.7	0.7899	0.9663
t+14	2503.0998	1033.5	1367.8702	624.4	0.7906	0.9643
t+15	2508.0793	1071.8	1364.9104	637.5	0.7898	0.9616
t+16	2511.8131	1129.4	1366.4333	663.0	0.7892	0.9574
t+17	2506.1916	1177.0	1352.6855	695.1	0.7901	0.9537
t+18	2502.3284	1226.3	1331.1768	718.3	0.7901	0.9496
t+19	2499.1290	1326.4	1305.9974	760.5	0.7890	0.9406
t+20	2498.5868	1447.5	1307.7390	826.3	0.7865	0.9284
t+21	2502.2396	1579.2	1305.1099	893.2	0.7826	0.9134
t+22	2499.7526	1738.9	1296.6306	973.1	0.7801	0.8936
t+23	2502.0577	1907.3	1288.8361	1061.2	0.7777	0.8708
t+24	2530.4582	2078.1	1293.3795	1148.1	0.7720	0.8463
Aggregate	2382.5835	1121.8	1258.3502	617.6	0.8078	0.9574

TABLE I: Tabular comparison of RMSE, MAE, and R^2 metrics between baseline LSTM model and GA-VMD-LSTM Ensemble predictions against actual test set values. Both models used lookbacks of 48h to predict the coming 24 hours.

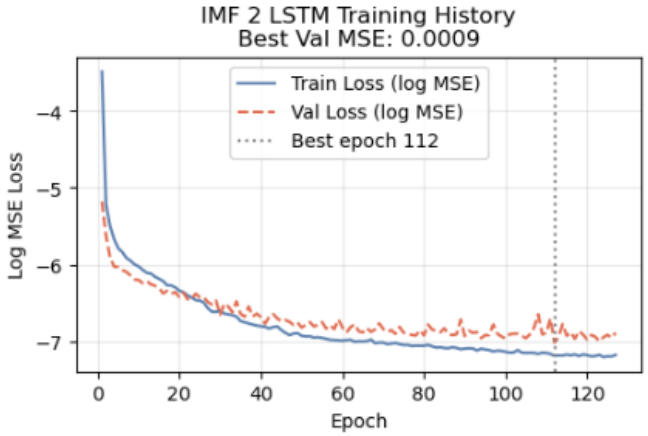


FIG. 3: Training and validation MSE loss curves per epoch for $K = 2$ IMF's LSTM, from the GA-VMD-LSTM. Loss is plotted on a log scale. Vertical line at epoch 112 shows the best model checkpoint. Modelling is of station SQ19.

IMF K	RMSE	R^2
IMF 1	440.5	0.9073
IMF 2	547.7	0.9837
IMF 3	468.3	0.9640
IMF 4	381.9	0.8444
IMF 5	384.2	0.4685
IMF 6	345.4	0.2239
IMF 7	315.7	0.2039
IMF 8	71.1	0.1290
Ensemble	1121.8	0.9574

TABLE II: Per-IMF RMSE and R^2 metrics between LSTM predictions and actual test values for the 48h lookback / 24h horizon GA-VMD-LSTM Ensemble.

oscillations which partially cancel upon summation.

Lastly, the effects of varying the lookback for various fixed horizons was investigated for both models. Figure 4 shows that for short term horizons ($h \leq 12$h), increasing lookback beyond a full diurnal cycle has little-to-no effect on predictions. For longer horizons of $h = 24$h, increasing lookback to a week caused a measurable drop in forecasting quality beyond $t + 18$, as the longer input sequence dilutes the model's sensitivity to the most recent and predictive weather data. For this horizon, longer lookbacks offered no benefit over 48h — thus suggesting that additional historical context carries diminishing informational value. The $h = 48$h test demonstrates the limited forecasting power weather data can grant: R^2 visibly plateaus after $t + 20$, suggesting the GA-VMD-LSTM reverts towards mean prediction when forecast horizon exceeds the predictive range of available meteorological data. A similar plot using RMSE instead of R^2 and those for the baseline LSTM can be found in the appendix §7.2.

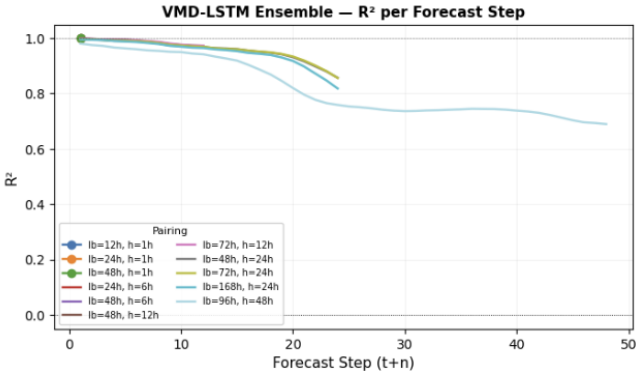


FIG. 4: Investigation of the effects of various (lookback, horizon) pairings on the GA-VMD-LSTM Ensemble's R^2 performance per predictive step. Modelling is of station SQ19.

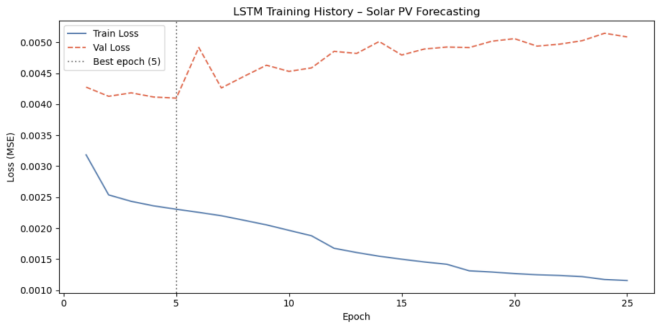


FIG. 5: Training and validation MSE loss curves per epoch for the Generalised Multi-Station LSTM. Loss is plotted on a log scale. Vertical line at epoch 5 shows the best model checkpoint.

3.2. Generalised LSTM

The generalised LSTM achieved an overall $R^2 = 0.9603$ and capacity scaled RMSE$= 0.0368$ for next-hour forecasting across 57 stations (Table III, appendix §7.3). Per-station R^2 ranged from 0.8062 to 0.9922. The weakest performer, S_H_Ho_Sports_Hall, had notably fewer training sequences (1542 vs. ∼2422 for most stations), suggesting data scarcity rather than architectural failure. The SQ station cluster consistently achieved $R^2 > 0.988$, reflecting their physical homogeneity. With such a large sample of similar stations, it is no surprise the model performed strongest.

Validation loss convergence was weak (Figure 5) reflecting the same struggle as the baseline single-station LSTM. Without VMD decomposition, the LSTM struggles to effectively map the multiple nonlinear correlations at play — a problem worsened by the large number of heterogenous PV stations.

4. CONCLUSION

This project implemented and evaluated three LSTM models. The single-station baseline LSTM struggled beyond near-term extrapolation due to the complex multi-timescale dynamics of PV generation. Decomposing this signal into intrinsic mode functions, as with the GA-VMD-LSTM Ensemble, showed strong evidence that physically dividing the task into simpler sub-problems yields superior results: a 53% and 51% reduction in RMSE and MAE from the baseline LSTM, and an R^2 improvement from 0.8078 to 0.9574 on average over the full 24-hour predictive horizons.

Lastly, the multi-station generalised LSTM showed the same pitfalls as the baseline: weak loss convergence demonstrated inability to learn the complex PV dynamics. A natural extension would be to perform VMD independently per station, with station-specific GA-optimised (K, α), and then train a shared multi-station ensemble on the first N low-frequency IMFs while aggregating the remaining high-frequency modes into a single residual channel.

5. USE OF GENERATIVE AI

Perplexity Research and Claude Sonnet 4.6 Thinking were used to generate boilerplate code. Instances of AI usage are outlined in the notebook, and accompanied with the respective prompt. All code was reviewed and tested before application.

6. REFERENCES

[1] Pan, H., Li, C., Xiao, F., Zhou, H. & Zhu, B. A pso-vmd-lstm-based photovoltaic power forecasting model incorporating pv converter characteristics. *Applied Sciences* **15** (2025). URL https://www.mdpi.com/2076-3417/15/19/10612.

[2] Hao, Y. *et al.* F10.7 daily forecast using lstm combined with vmd method. *Space Weather* **22**.

7. APPENDIX

7.1. GA-VMD-LSTM per-IMF loss curves

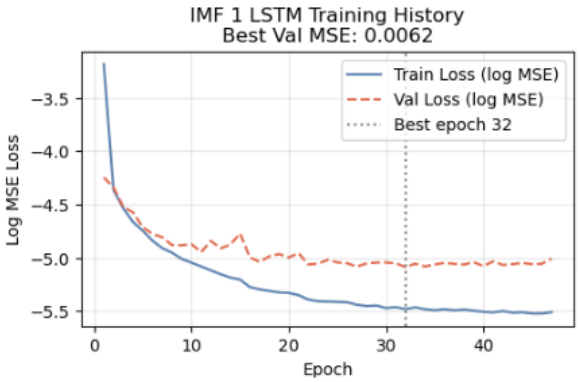


FIG. 6: Training and validation MSE loss curves per epoch for $K = 1$ IMF's LSTM, from the GA-VMD-LSTM. Loss is plotted on a log scale. Vertical line at epoch 32 shows the best model checkpoint.

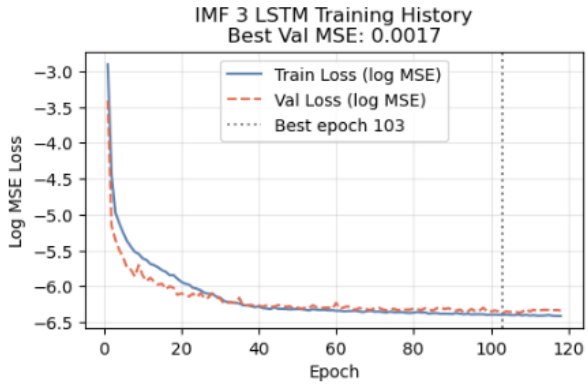


FIG. 7: Training and validation MSE loss curves per epoch for $K = 3$ IMF's LSTM, from the GA-VMD-LSTM. Loss is plotted on a log scale. Vertical line at epoch 103 shows the best model checkpoint.

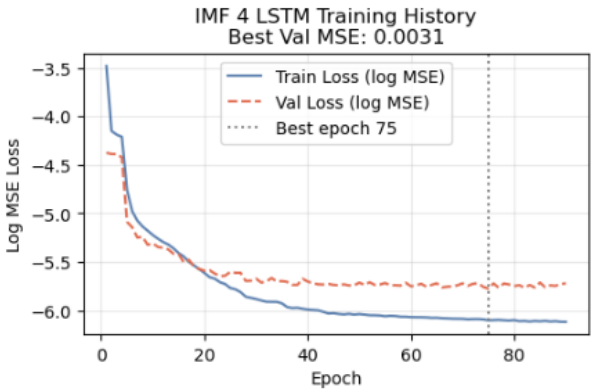


FIG. 8: Training and validation MSE loss curves per epoch for $K = 4$ IMF's LSTM, from the GA-VMD-LSTM. Loss is plotted on a log scale. Vertical line at epoch 75 shows the best model checkpoint.

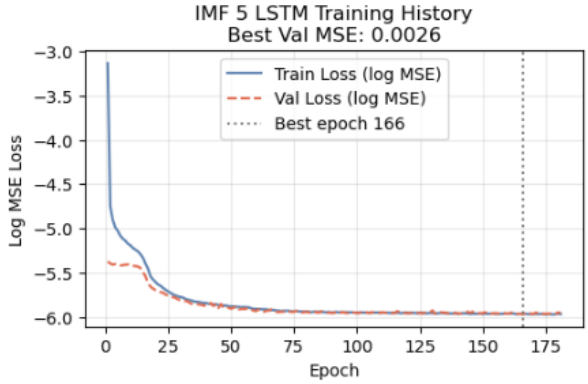


FIG. 9: Training and validation MSE loss curves per epoch for $K = 5$ IMF's LSTM, from the GA-VMD-LSTM. Loss is plotted on a log scale. Vertical line at epoch 166 shows the best model checkpoint.

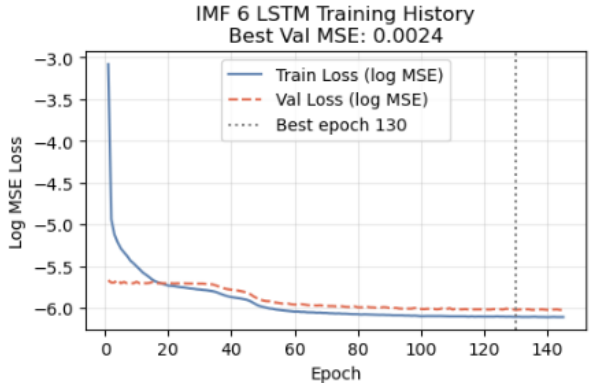


FIG. 10: Training and validation MSE loss curves per epoch for $K = 6$ IMF's LSTM, from the GA-VMD-LSTM. Loss is plotted on a log scale. Vertical line at epoch 130 shows the best model checkpoint.

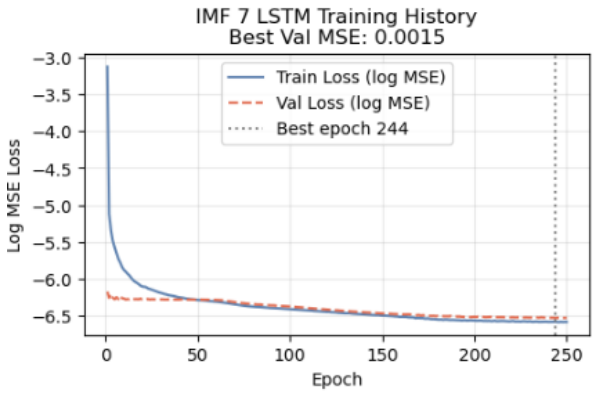


FIG. 11: Training and validation MSE loss curves per epoch for $K = 7$ IMF's LSTM, from the GA-VMD-LSTM. Loss is plotted on a log scale. Vertical line at epoch 244 shows the best model checkpoint.

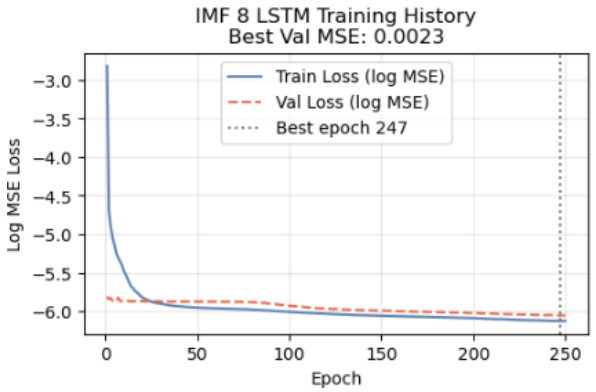


FIG. 12: Training and validation MSE loss curves per epoch for $K = 8$ IMF's LSTM, from the GA-VMD-LSTM. Loss is plotted on a log scale. Vertical line at epoch 247 shows the best model checkpoint.

7.2. Lookback-Horizon Pairing Investigation

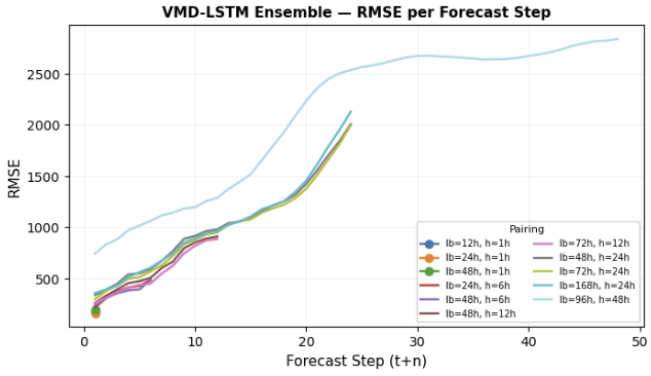


FIG. 13: Investigation of the effects of various (lookback, horizon) pairings on the GA-VMD-LSTM Ensemble's RMSE performance per predictive step.

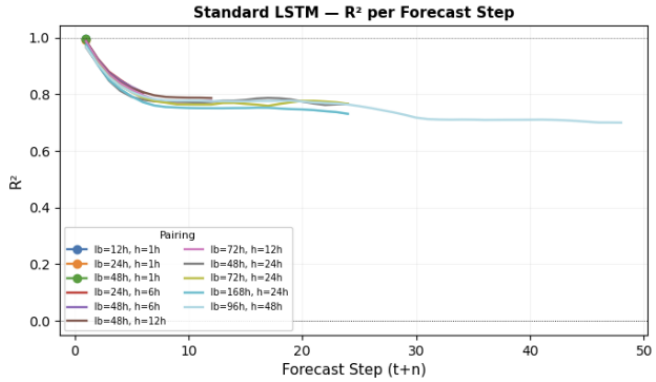


FIG. 14: Investigation of the effects of various (lookback, horizon) pairings on the baseline LSTM's R^2 performance per predictive step.

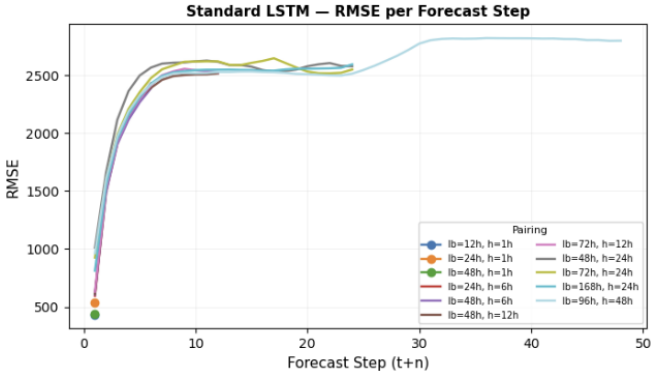


FIG. 15: Investigation of the effects of various (lookback, horizon) pairings on the baseline LSTM's RMSE performance per predictive step.

7.3. Full Generalised LSTM Next-Hour Forecast Results Table

Station	R^2	RMSE	Capacity (kW)	N_{seq}
S_H_Ho_Sports_Hall	0.8062	0.0354	42.00	1542
UG_Hall2_2F	0.8961	0.0245	23.40	2422
UG_Hall2_RF	0.9371	0.0376	8.00	2372
SQ_Block_S	0.9372	0.0385	32.85	2422
Zone_A1	0.9423	0.0385	33.30	2422
SQ_Apartment37_48	0.9435	0.0328	16.22	2367
SQ_Block_R	0.9442	0.0457	33.30	2422
SQ1	0.9449	0.0484	27.60	2422
Zone_A2	0.9461	0.0395	249.50	2396
Zone_L2	0.9464	0.0479	10.00	2346
SQ2	0.9465	0.0409	33.30	2422
UG_Hall7	0.9467	0.0275	31.98	2422
SQ_Apartment13_24	0.9469	0.0558	17.00	2392
SQ_Apartment25_36	0.9471	0.0487	17.00	2392
SQ_Apartment1_12	0.9476	0.0560	17.00	2392
SQ3	0.9479	0.0427	33.30	2422
Zone_A7	0.9480	0.0372	12.78	2397
Library	0.9482	0.0289	170.20	1852
Zone_A6	0.9483	0.0519	50.00	2422
UG_Hall3	0.9487	0.0456	25.00	2422
SQ4	0.9490	0.0433	33.30	2422
Zone_J1	0.9495	0.0415	25.55	2422
Zone_A5	0.9504	0.0435	17.00	2422
Zone_D	0.9509	0.0474	68.62	2422
Wong_Check_She_Research_Centre	0.9516	0.0489	100.00	2422
UG_Hall9	0.9518	0.0464	25.00	2422
Zone_A4	0.9521	0.0386	55.00	2422
UG_Hall8	0.9525	0.0414	25.00	2422
LSK_North	0.9539	0.0468	55.00	2422
UG_Hall4	0.9545	0.0377	32.76	2422
SQ567	0.9548	0.0386	47.97	2422
UG_Hall4_Flexible_PV	0.9552	0.0473	19.84	2255
Indoor_Sport_Center	0.9554	0.0494	33.30	2422
UG_Hall6	0.9555	0.0467	25.00	2422
LSK_South	0.9570	0.0418	25.00	1494
Zone_J2	0.9597	0.0297	50.00	2396
Zone_L1__Station_2_	0.9625	0.0412	25.00	2298
SQ_Block_Q	0.9690	0.0309	26.26	2239
SQ_Apartment37_48_Flexible_PV	0.9709	0.0478	3.72	1925
CYT__Station_1_	0.9733	0.0293	25.00	2102
UG_Hall7_Flexible_PV	0.9744	0.0112	5.58	2293
SQ_Block_Q__Phrase_2_	0.9752	0.0233	10.30	2277
CYT__Station_2_	0.9854	0.0244	25.00	2252
Zone_L1__Station_1_	0.9861	0.0239	25.00	2330
SQ17	0.9880	0.0178	37.05	2249
SQ9	0.9884	0.0174	30.81	2317
SQ16	0.9888	0.0171	33.15	2293
SQ10	0.9890	0.0167	30.40	2259
SQ18	0.9895	0.0141	33.93	2191
Tower_A	0.9899	0.0185	40.29	2293
SQ8	0.9905	0.0156	32.51	2294
SQ13	0.9913	0.0161	28.47	2235
SQ11	0.9919	0.0131	32.51	2260
SQ19	0.9921	0.0139	32.51	2293
SQ12	0.9921	0.0137	32.12	2284
SQ14	0.9922	0.0140	32.90	2226
SQ15	0.9922	0.0124	33.29	2284
Overall	0.9603	0.0368	-	131493

TABLE III: Per-station R^2, capacity-normalised RMSE, capacity, and number of training sequences for the generalised multi-station LSTM for next-hour forecasting. Sorted by ascending R^2.